

MADHULEKA BHOWMICK

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Education

Ramrao Adik Institute of Technology, DY Patil University

B.Tech in Information Technology

2022 – 2026

Navi Mumbai, India

Experience

Infocloud IT Services Pvt Ltd

AI Intern

Feb 2026 – Mar 2026

Mumbai, India

- Contributed to development of a **visitor management platform** for a **UAE client**, integrating **facial recognition** and **smart card scanning** to automate visitor verification and surface real-time meeting and conference booking status.
- Designed and implemented an **NLP-based AI chatbot** with an FAQ database; engineered backend APIs using **FastAPI**, built databases with **Prisma + PostgreSQL**, tested endpoints via **Postman**, deployed using **Nginx**, and proposed a **text-to-speech feature** to improve accessibility.

Projects

Product Recommender Web App – CI/CD | [Live](#) | [GitHub](#) | *React, REST, Vercel*

Nov 2025

- Developed a **full-stack product recommender web application** using **React** and **REST APIs** to provide automated product suggestions to users.
- Implemented **GitHub-based CI/CD pipeline** using **Vercel** to automate deployments and resolved build/runtime issues, improving deployment reliability.

AI Legal Assistant (BNS) | *Python, Flask, NLP, LLMs*

May 2025

- Built an **AI-powered legal assistant** using **Flan-T5 LLM** for legal text summarization and **FAISS vector search** for retrieving relevant case laws.
- Developed secure **Flask REST APIs** enabling lawyers and public users to query legal information efficiently.

Lung Cancer Detection (PET Scans) | [GitHub](#) | *Deep Learning, Computer Vision*

Jan 2025

- Trained **deep learning computer vision models** on a dataset of **10,000+ PET scan images** to detect lung cancer patterns.
- Achieved **80% precision** in medical image classification and presented research findings at an **international conference (2025)**.

SafeSpeak – Hate Speech Detection | [GitHub](#) | *Python, Flask, NLP*

Aug 2024

- Developed **SafeSpeak**, a hate-speech detection system using **NLP** and **Logistic Regression** trained on **24,000+ tweets**.
- Achieved **94% classification accuracy** and deployed the model using **Flask REST APIs** for real-time text moderation.

Technical Skills

- Programming:** Python, SQL, HTML, CSS
- Machine Learning & NLP:** Logistic Regression, Text Classification, LLM Integration (Flan-T5), Vector Search (FAISS)
- Frameworks:** Flask, REST APIs
- Tools & Cloud:** Git, GitHub, VS Code, AWS (EC2, S3)
- Cybersecurity:** Capture The Flag (CTF) Challenges – TCS HackQuest

Achievements & Publications

- AI Legal Assistant (BNS)** — Accepted at **IC3ET 2026 IEEE International Conference** (Record #64989).
- TCS HackQuest Season 10** — Qualified for **Round 2 (Cyber Warrior)** among national participants.
- Ranked **5th out of 150+ participants** in **Clash of Codes 2.0 Hackathon**.
- Solved all challenges in the **MaTPO Programming Idol 2025**.
- Qualified for the **National AI Olympiad 2025**.

Certifications

- AWS Cloud Architecting** – EC2, S3
- IBM iX** – LLMs and Foundation Models
- Microsoft** – ML Models in Python
- CodeChef Bronze Medal** – Problem Solving